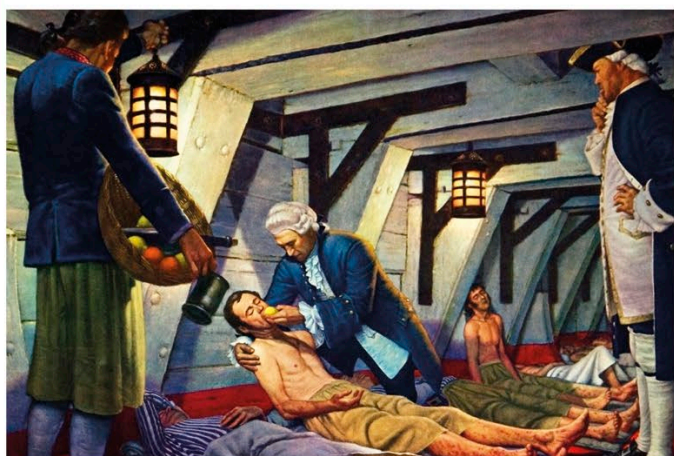


Developing treatments

Advances in medical science in the past 250 years have led to the development of a wide range of effective treatments for controlling or even eliminating diseases. Many have come about through trial and error, with innovation often racing ahead of scientific understanding.



20th-century painting of James Lind giving a lemon to a scurvy patient

Pump handle for extracting and injecting blood

Needle is attached to the pump.



Blundell's blood transfusion apparatus, 19th century

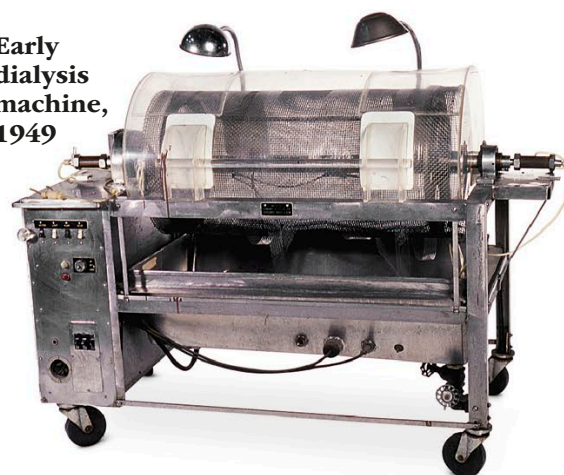
BLOOD TRANSFUSION

The British doctor James Blundell performed the world's first human blood transfusion in 1818. He injected a woman who had lost blood during childbirth with blood drawn from her husband. Although Blundell's procedure was successful, transfusions at the time didn't always work—blood types weren't fully understood and mixing different blood types could prove fatal.

PREVENTING SCURVY

Until the 18th century, sailors on long journeys often came down with a mysterious illness, which we now know to be scurvy. This is caused by a lack of vitamin C, which is present in citrus fruits. In 1747, the British surgeon James Lind demonstrated that sailors who regularly ate citrus fruits, such as lemons, didn't suffer from this illness. When the British navy finally took his advice, it cured scurvy in sailors almost overnight.

Early dialysis machine, 1949



KIDNEY DIALYSIS

In 1944, the Dutch doctor Willem Kolff invented a dialysis machine, which treats patients whose kidneys have failed. This first "artificial kidney" was a bulky device that cleaned the patient's blood of waste products by taking it out of the body, filtering it through a membrane, and then returning it. Today, this procedure—kidney dialysis—is fairly commonplace.

FAST FACTS

- The membrane in Willem Kolff's original kidney dialysis machine was made of artificial sausage skin.
- Blood types were discovered in 1901 by the Austrian physician Karl Landsteiner.
- The New Zealand surgeon Harold Gillies developed methods for repairing the facial injuries of soldiers in World War I that led to him being called the "father of plastic surgery."
- Today, laser eye surgery takes just a few minutes to perform.